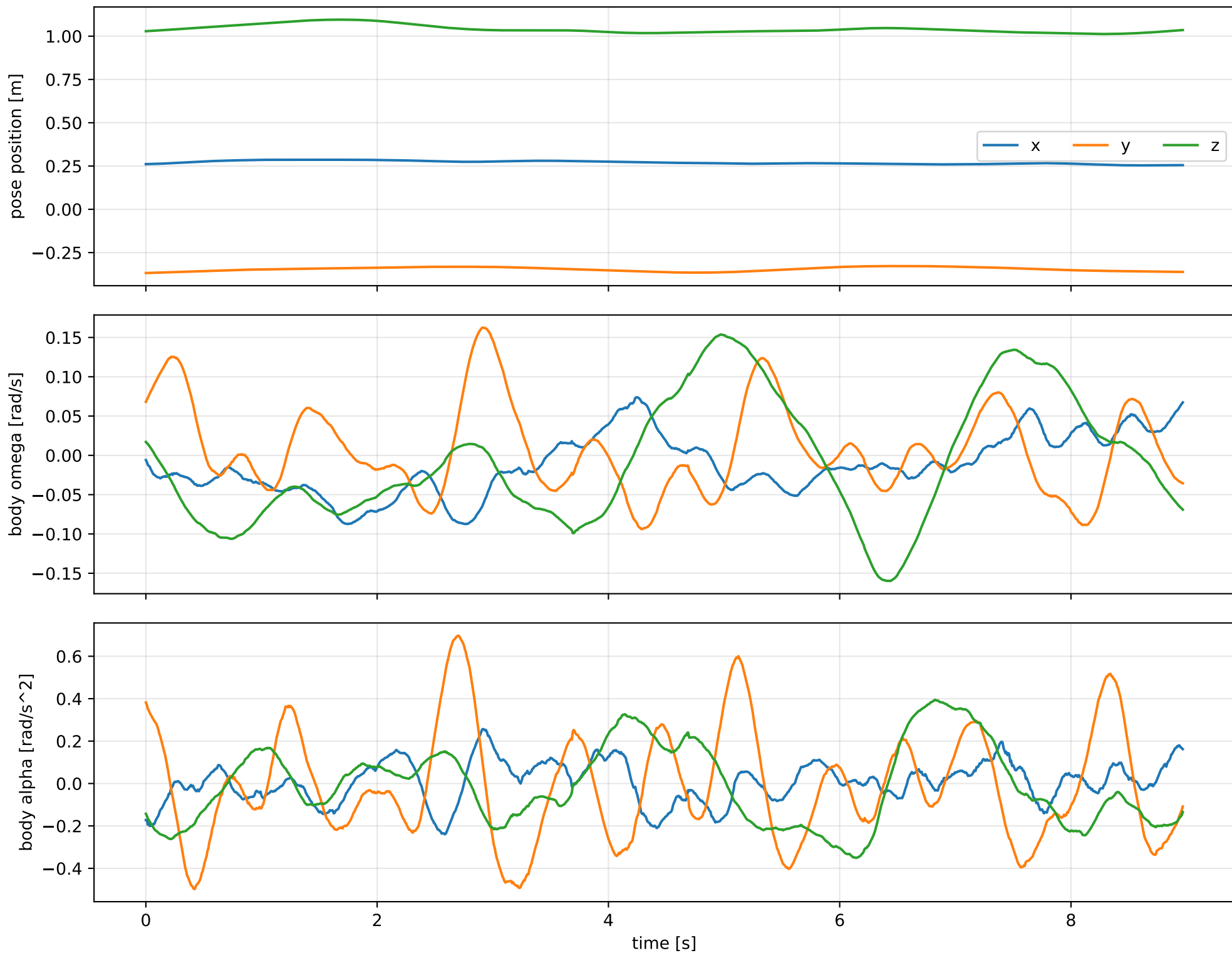
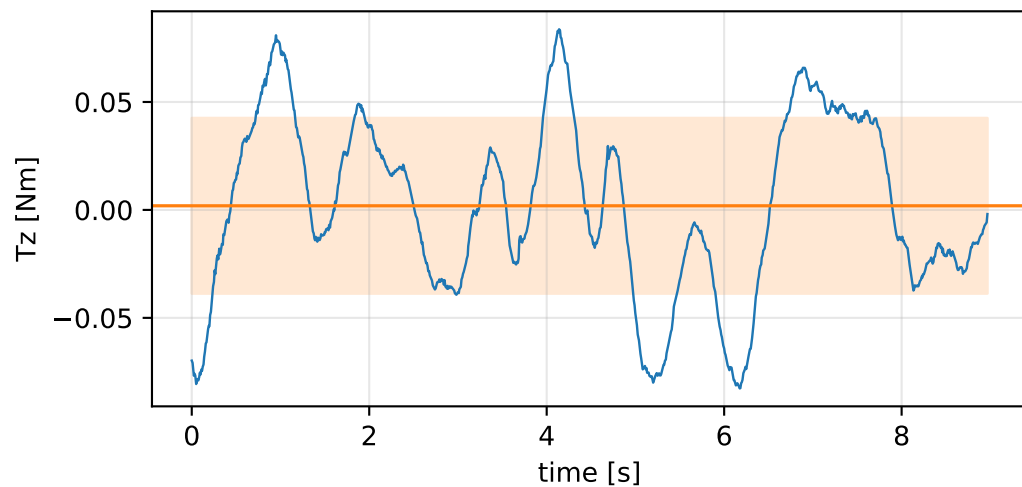
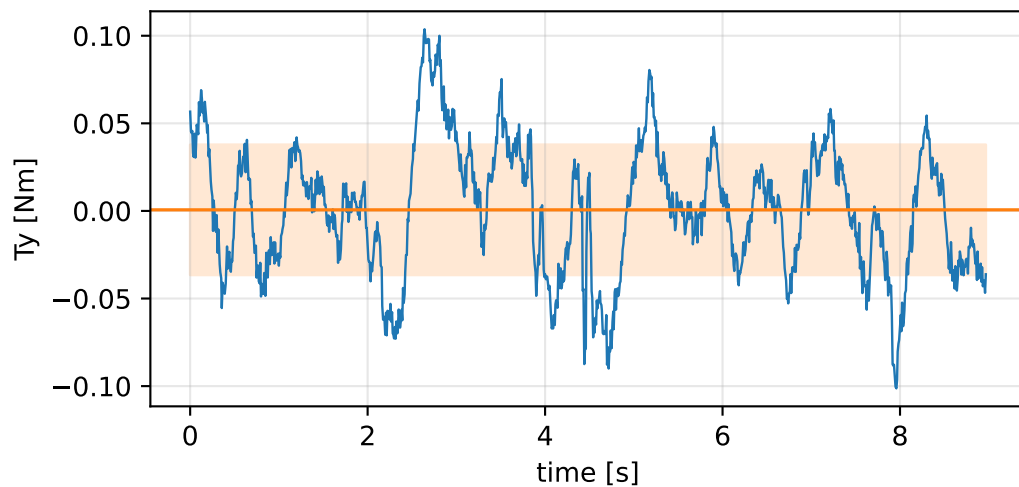
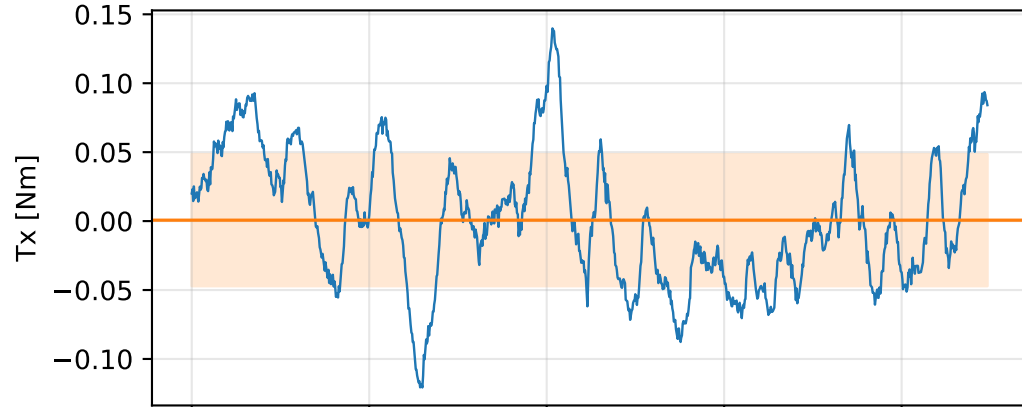
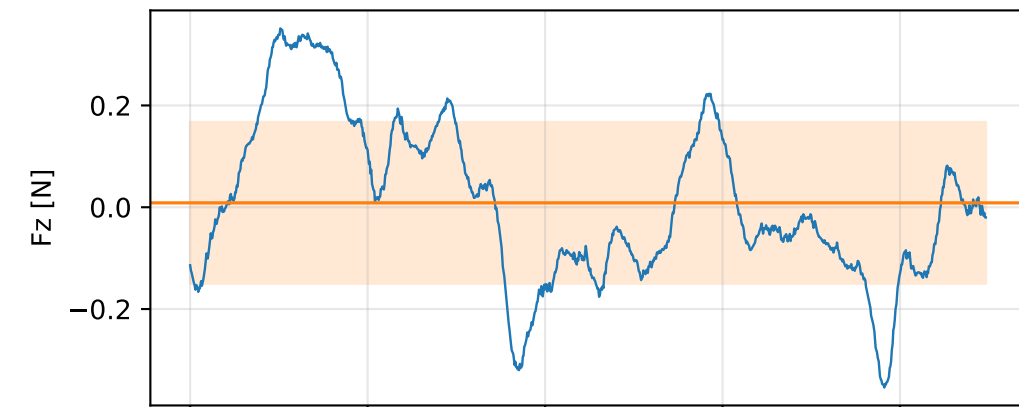
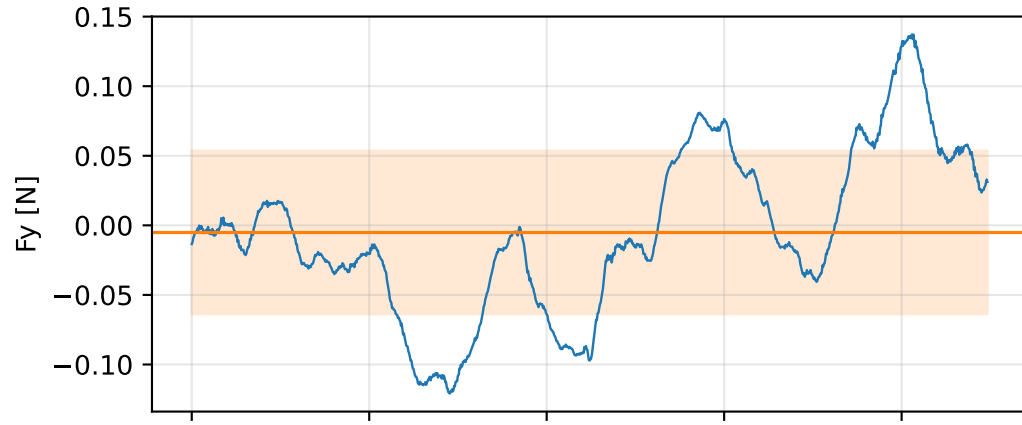
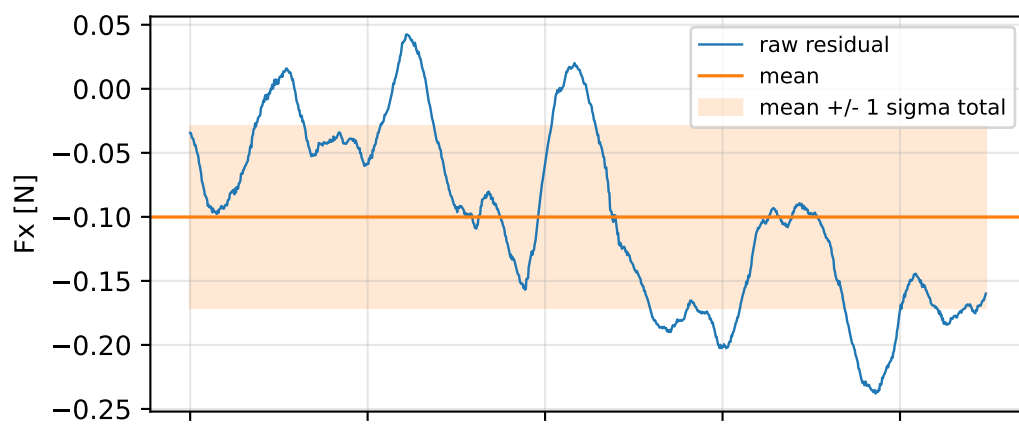


default: SG trajectory and derived motion



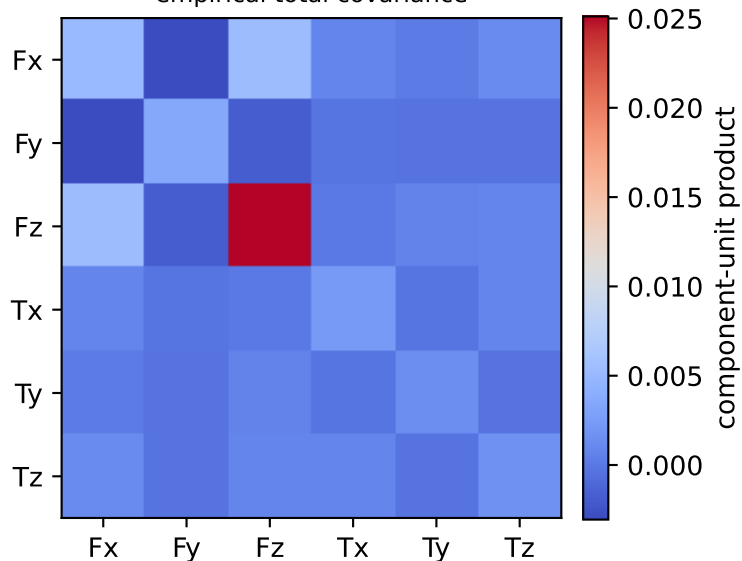
default: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



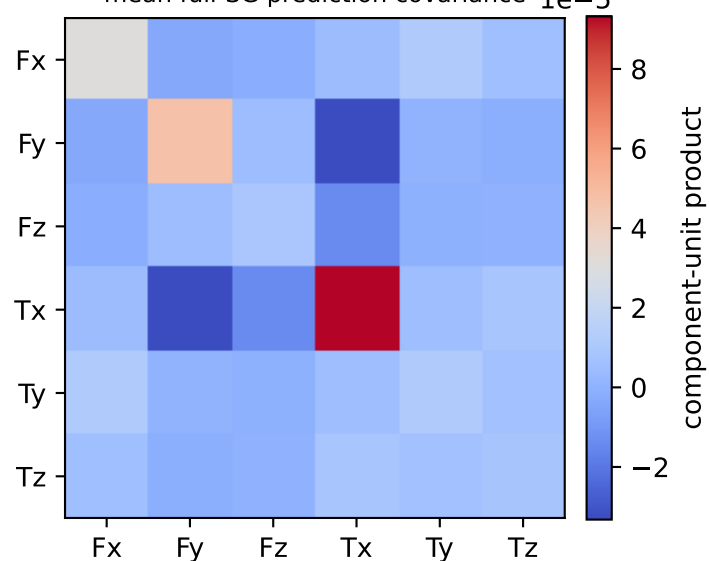
vector RMS about zero: force 0.20897 N, torque 0.072924 Nm; component std about mean: force [0.0709 0.0586 0.1585], torque [0.0478 0.0372 0.0406]

default: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.001 0.001 0.003 0.006 0.027]

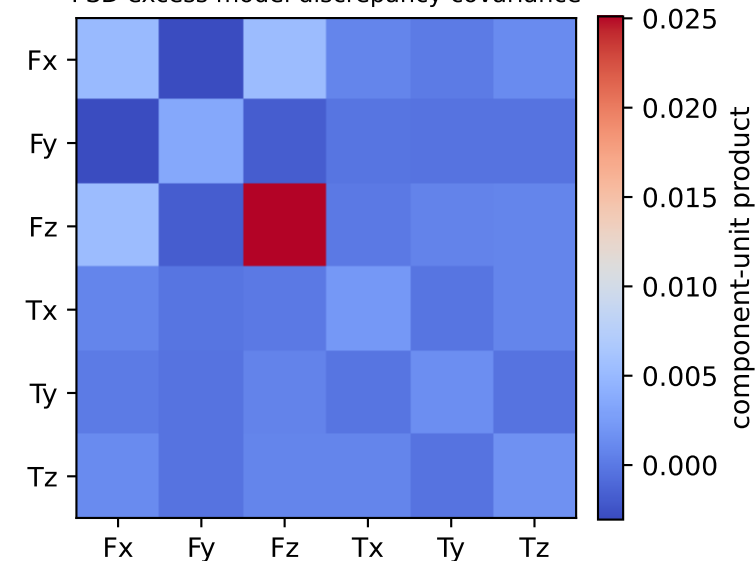
empirical total covariance



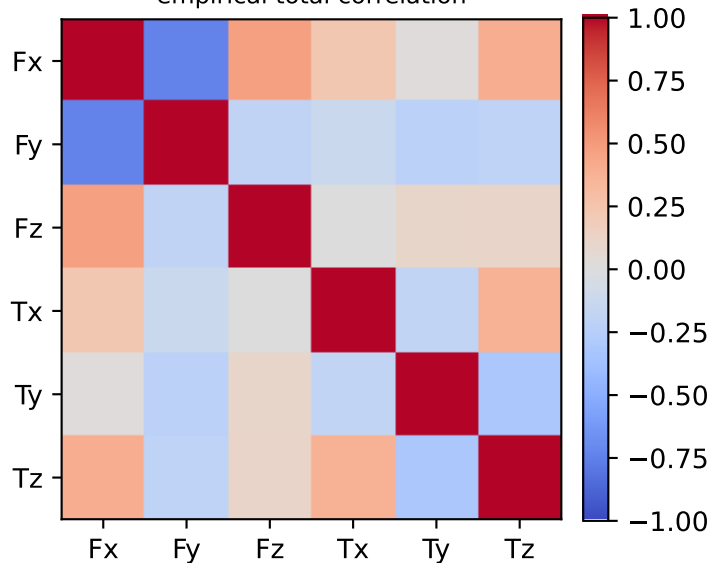
mean full-SG prediction covariance $1e-5$



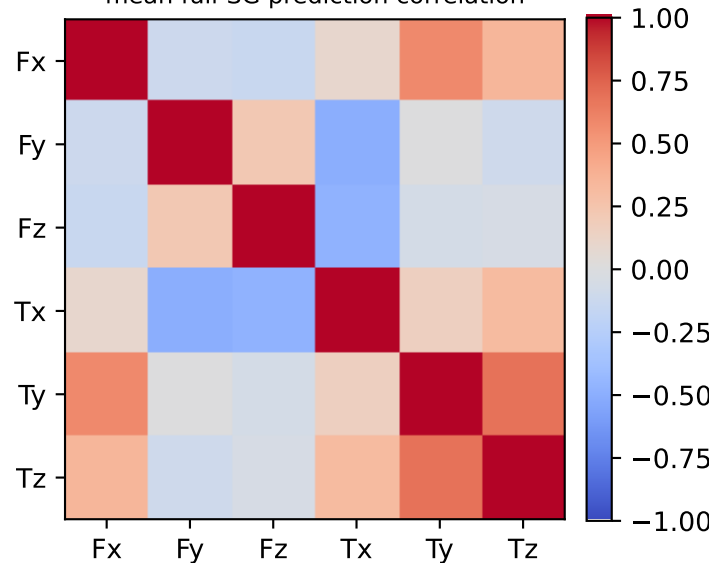
PSD excess model discrepancy covariance



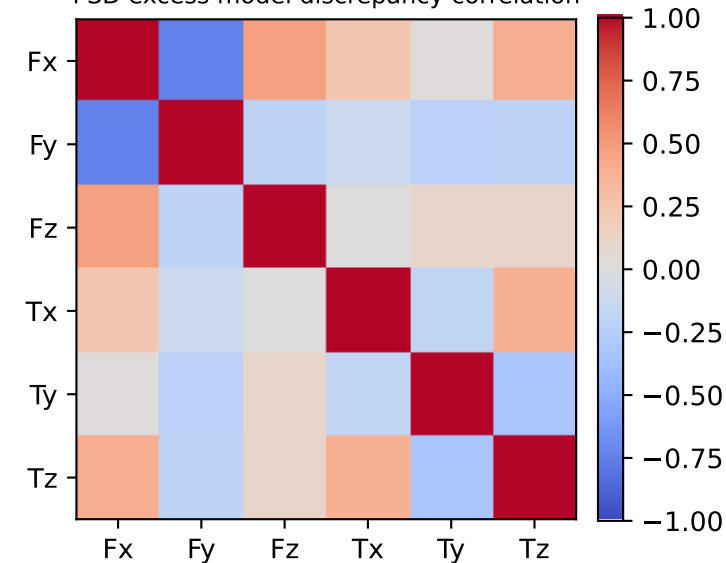
empirical total correlation



mean full-SG prediction correlation

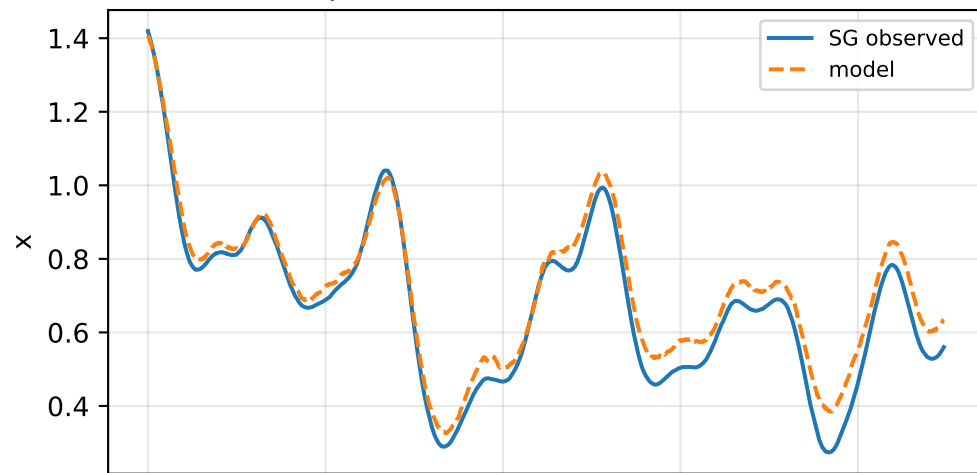


PSD excess model discrepancy correlation

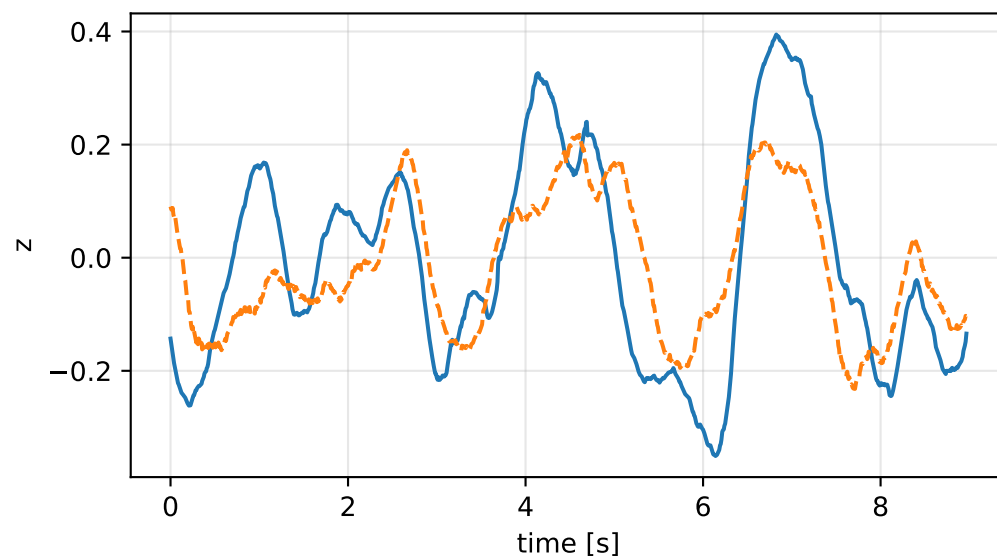
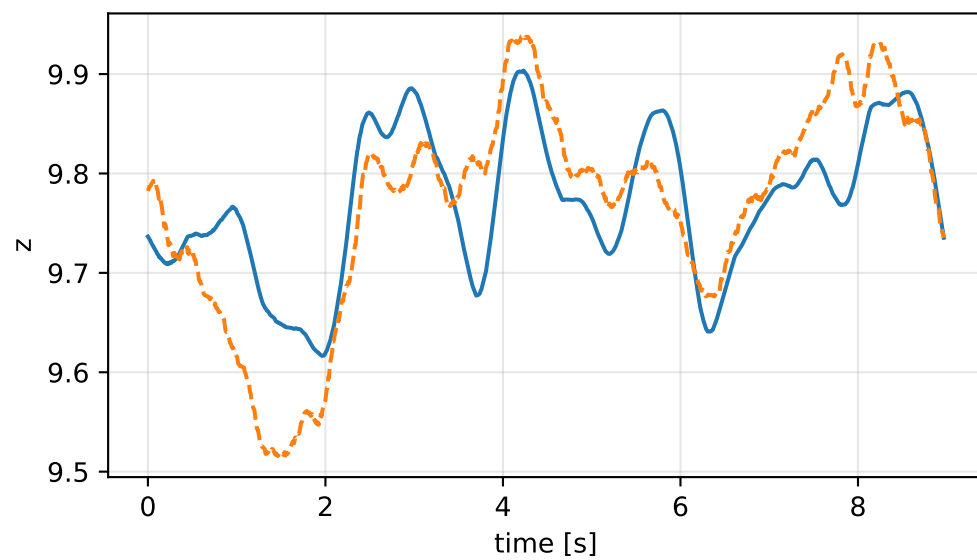
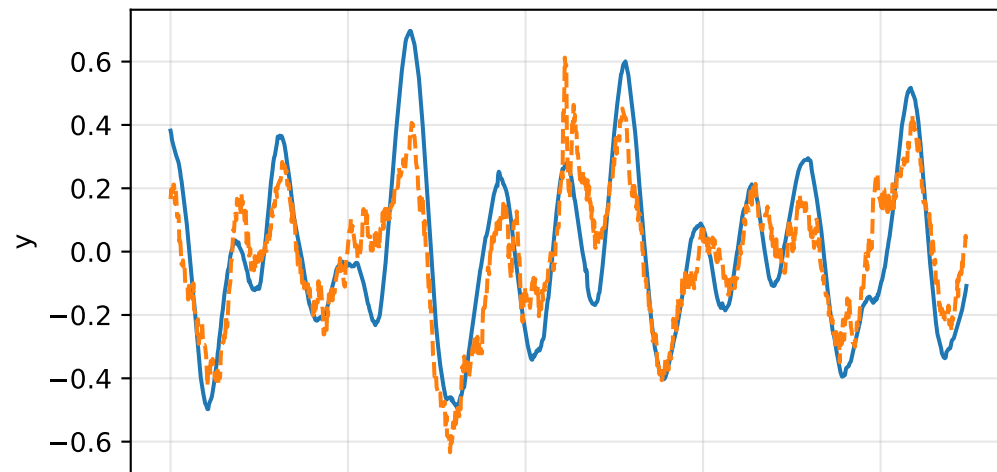
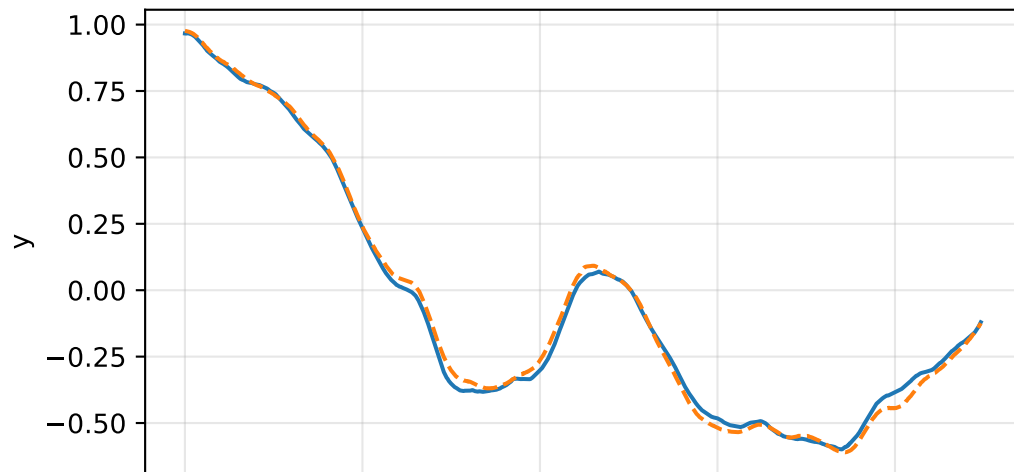
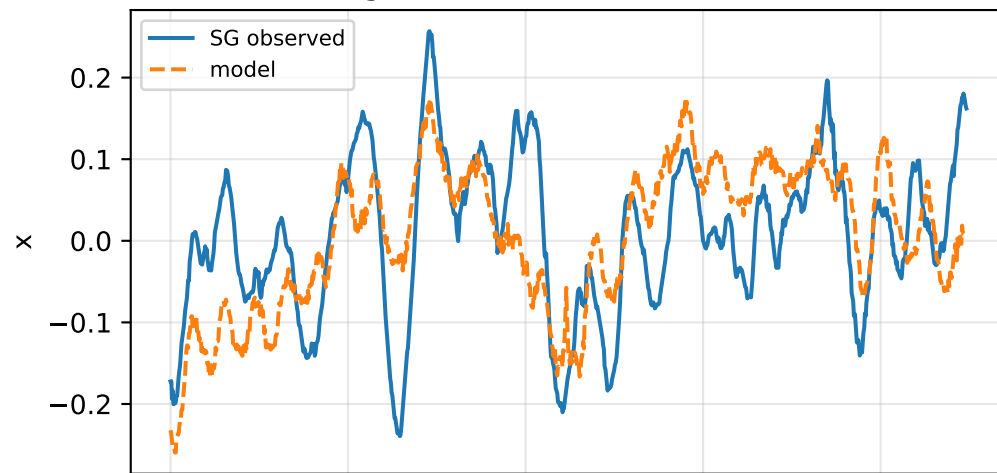


default: acceleration residual objective

specific acceleration [m/s²]

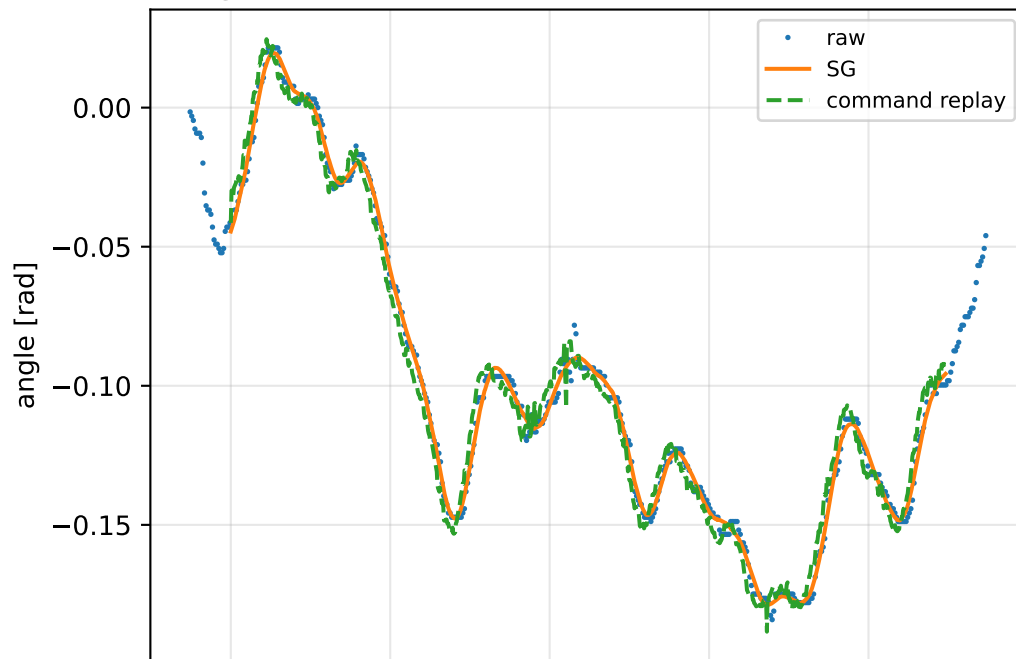


angular acceleration [rad/s²]

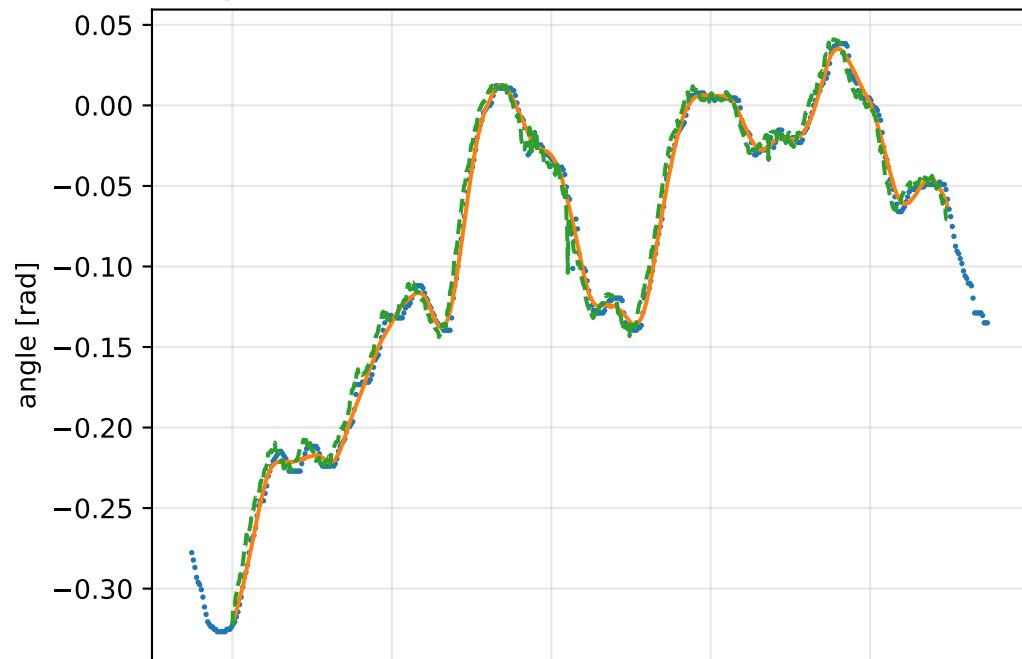


default: gimbal measurement audit (objective=measured_sg)

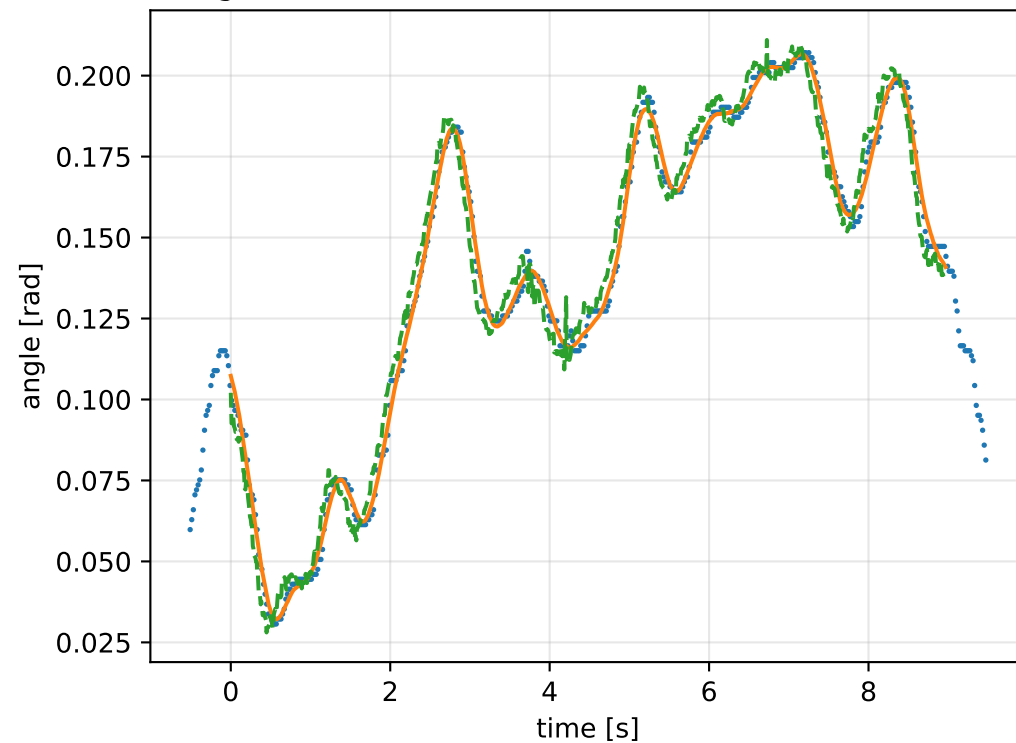
gimbal 1: RMSE=0.006345, max=0.01494 rad



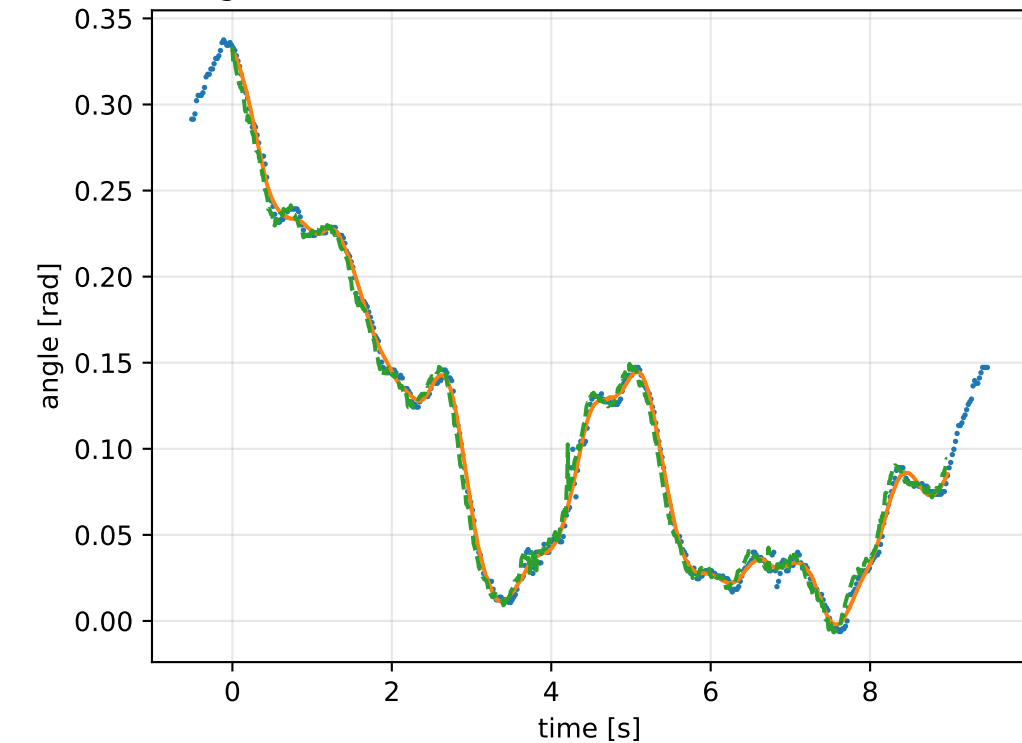
gimbal 2: RMSE=0.008835, max=0.04462 rad



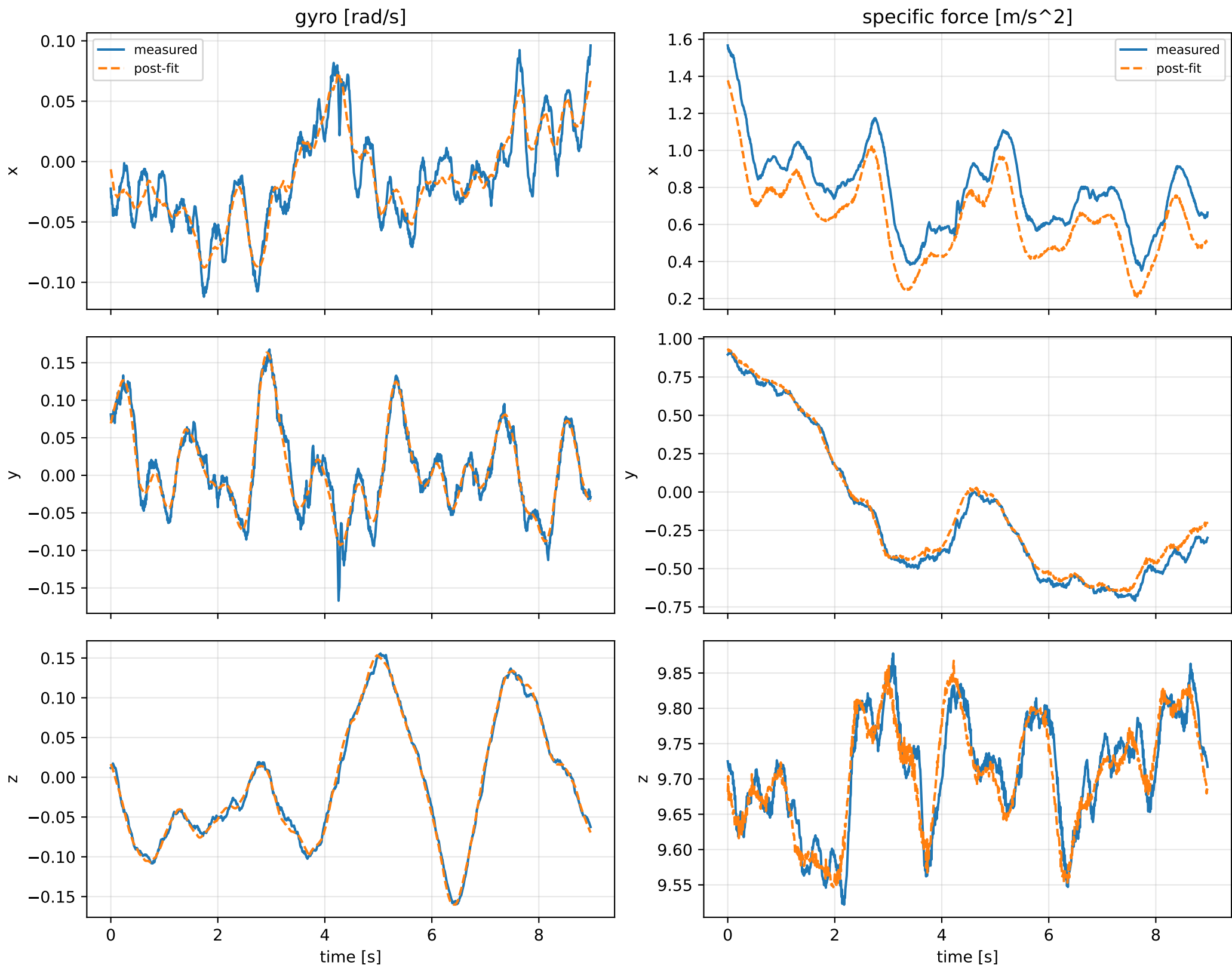
gimbal 3: RMSE=0.007019, max=0.01611 rad



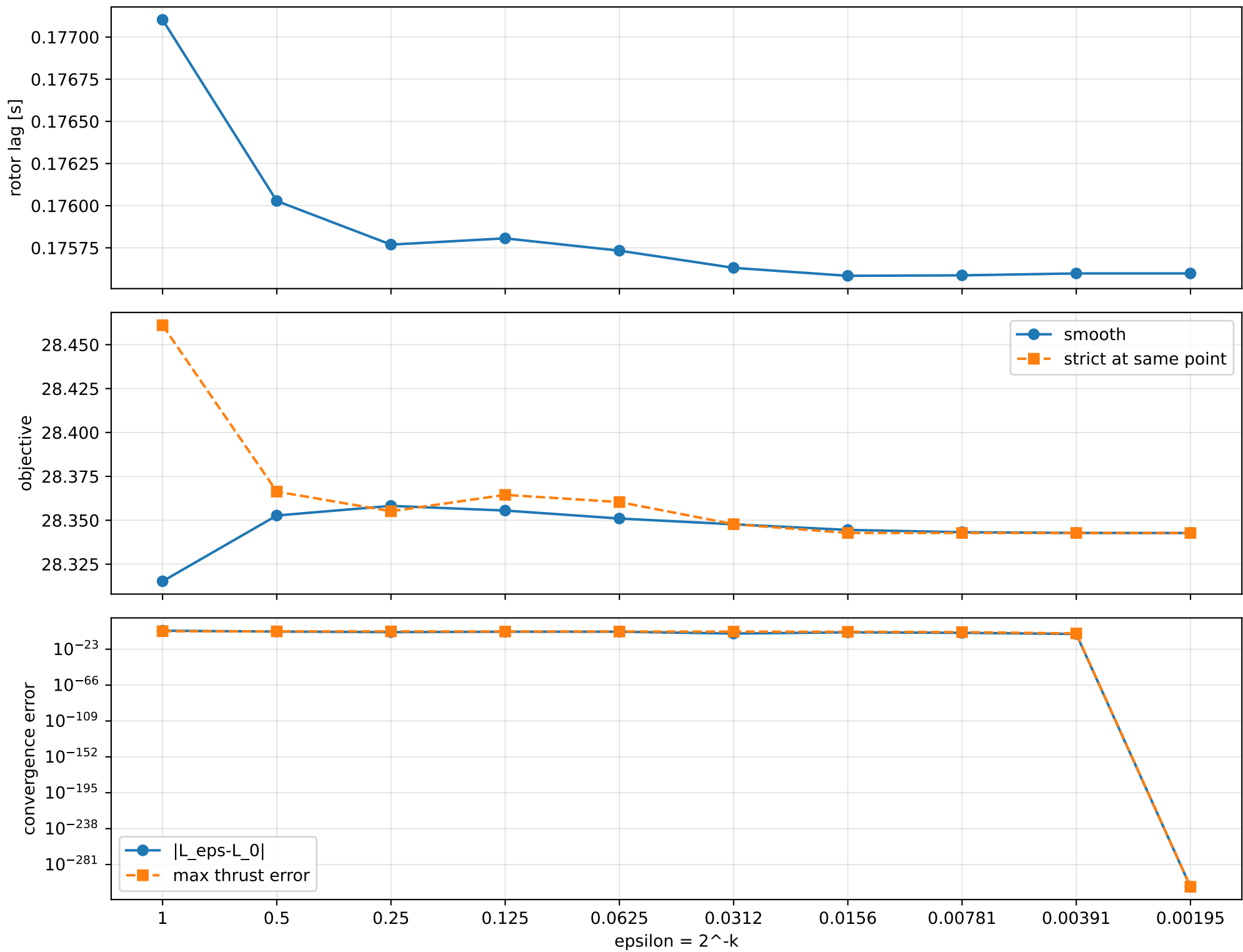
gimbal 4: RMSE=0.006321, max=0.03642 rad



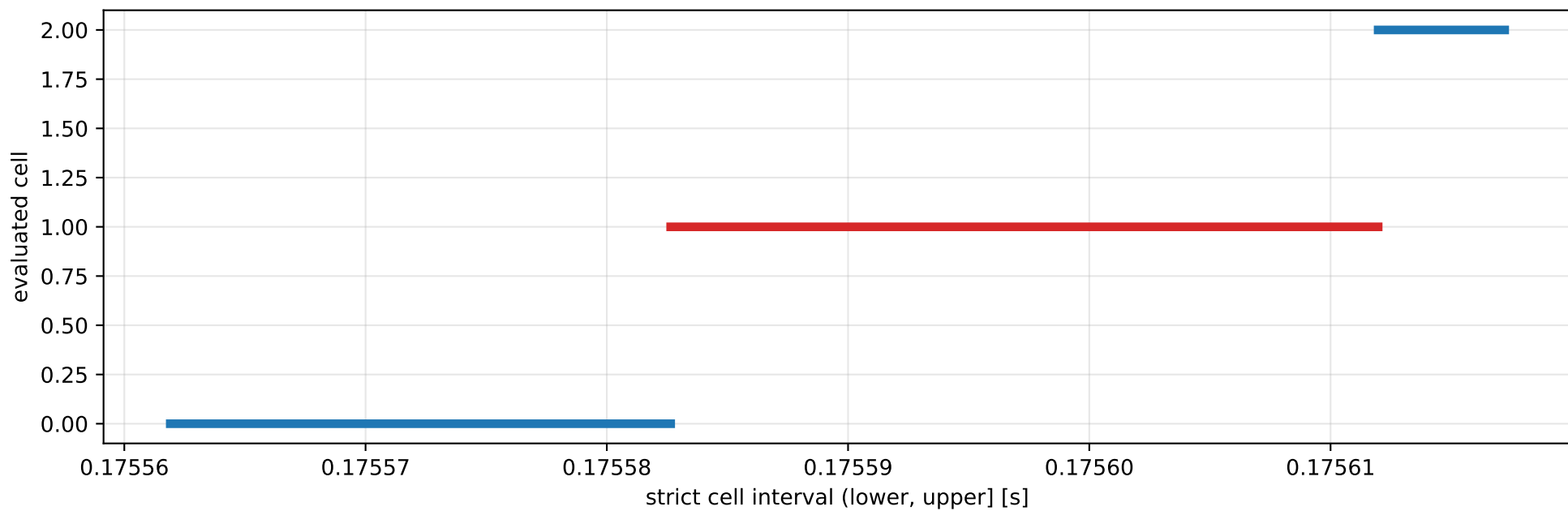
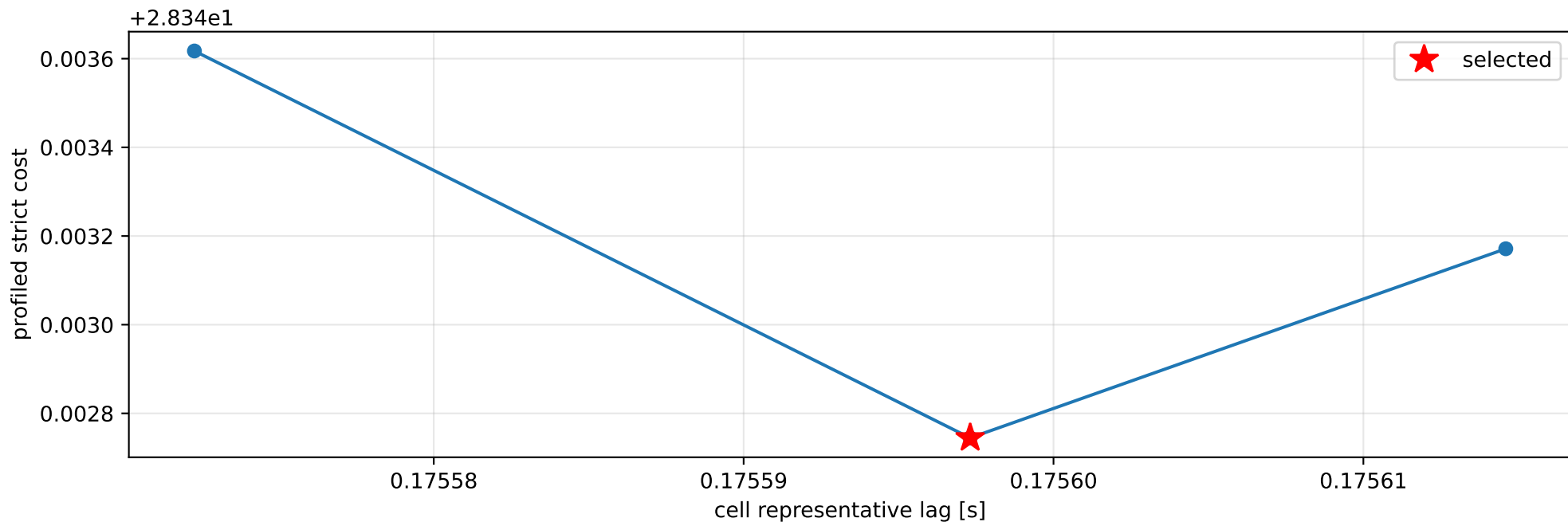
default: IMU comparison (diagnostic only)



default: smooth-to-strict continuation



default: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175597984 s; data boundary=False

default: parameter estimate

Case: default

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 5.95395056

inertia [kg m²]:

```
[[ 1.38088917 -0.03064693  0.1074512 ]  
 [-0.03064693  0.65296994 -0.01821479]  
 [ 0.1074512  -0.01821479  0.76743356]]
```

CoG body [m]: [-0.01982262 -0.05513055 -0.00503056]

force effectiveness: [2.64671275 2.58999662 1.36647123 1.65924686]

scale-free J/m [m²]:

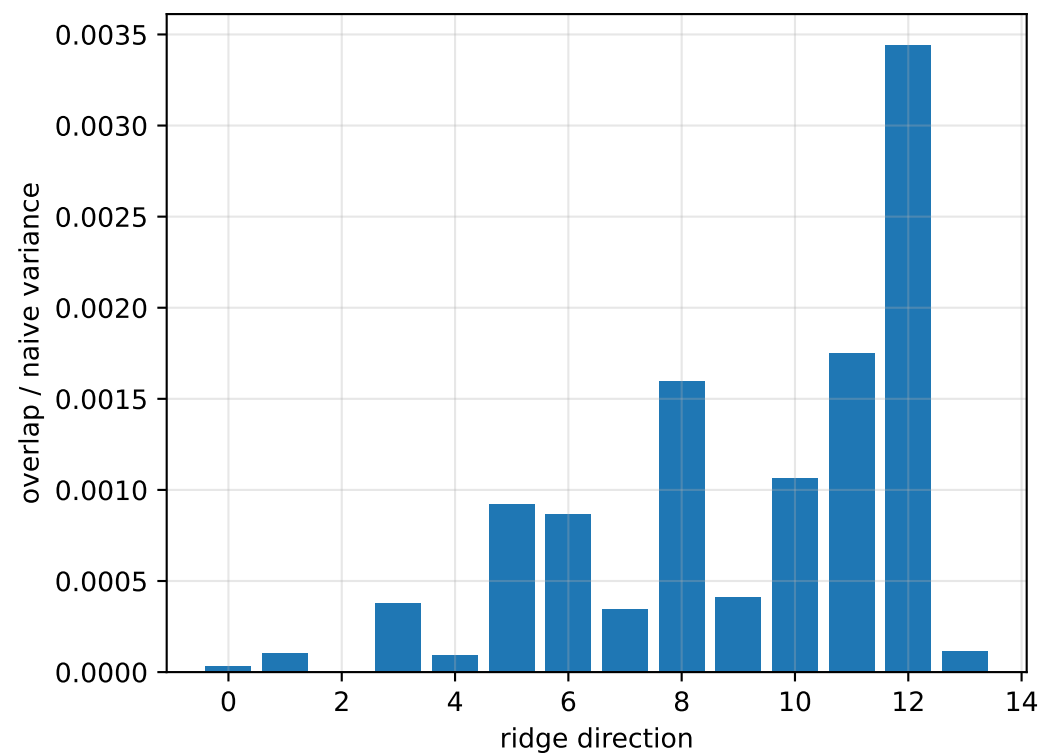
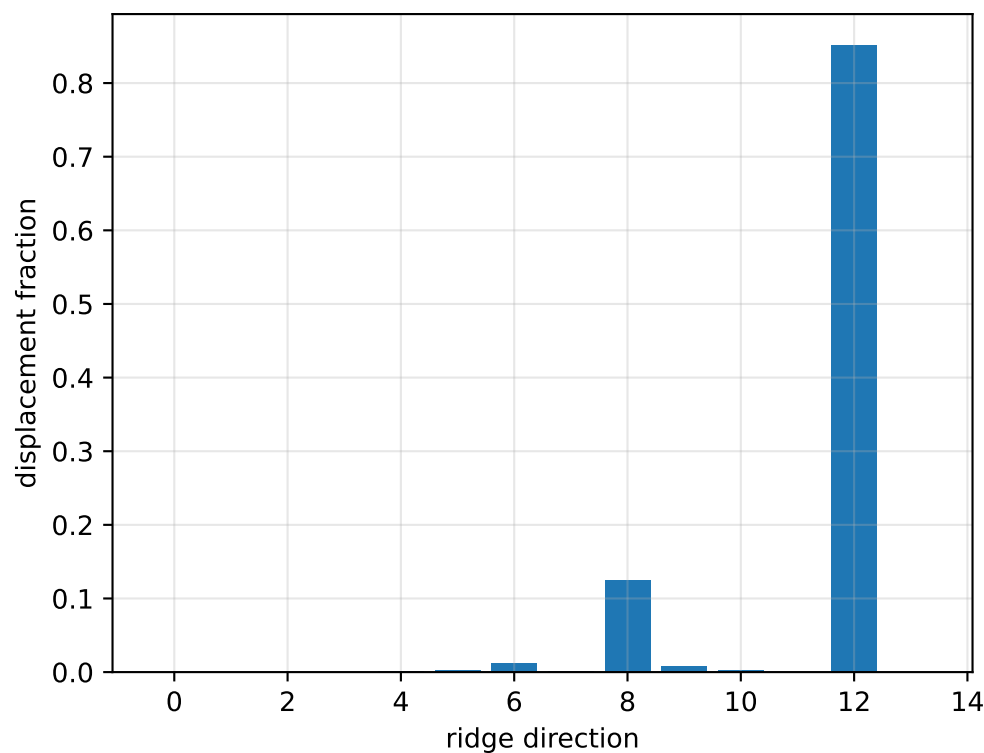
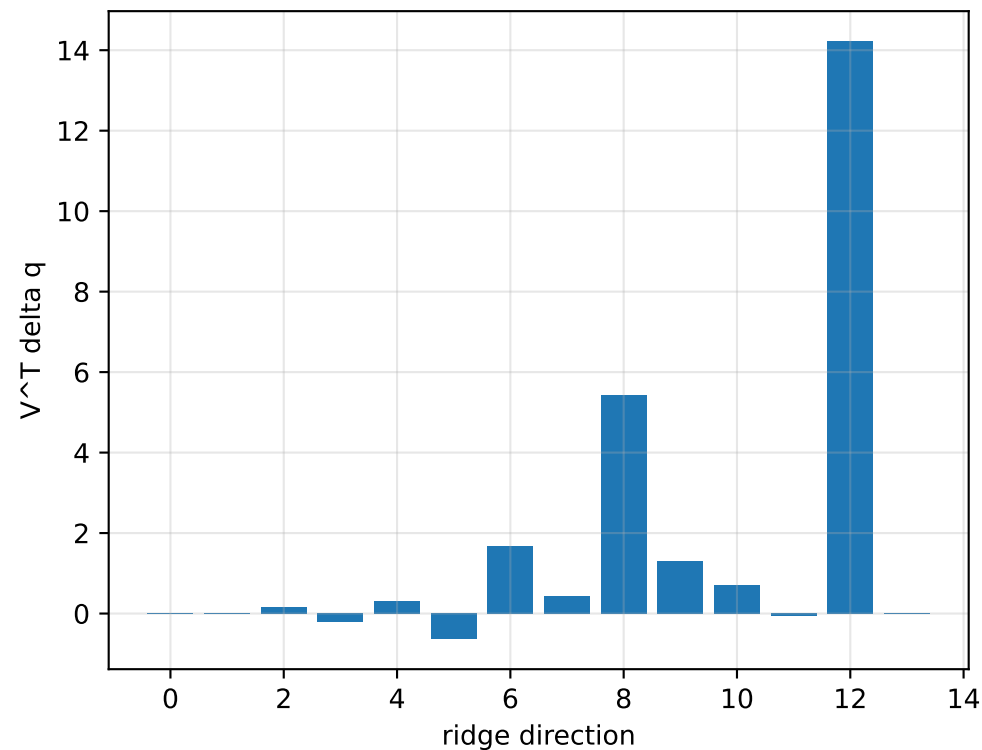
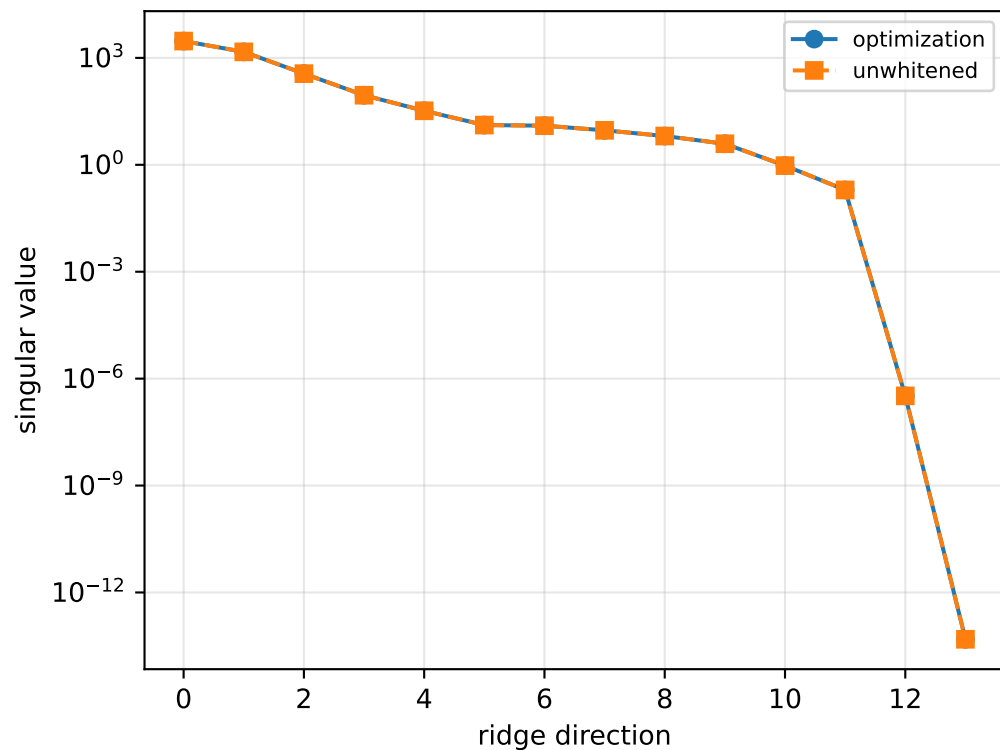
```
[[ 0.23192822 -0.00514733  0.01804704]  
 [-0.00514733  0.10967003 -0.00305928]  
 [ 0.01804704 -0.00305928  0.12889485]]
```

scale-free f/m: [0.44453052 0.43500472 0.22950665 0.27867999]

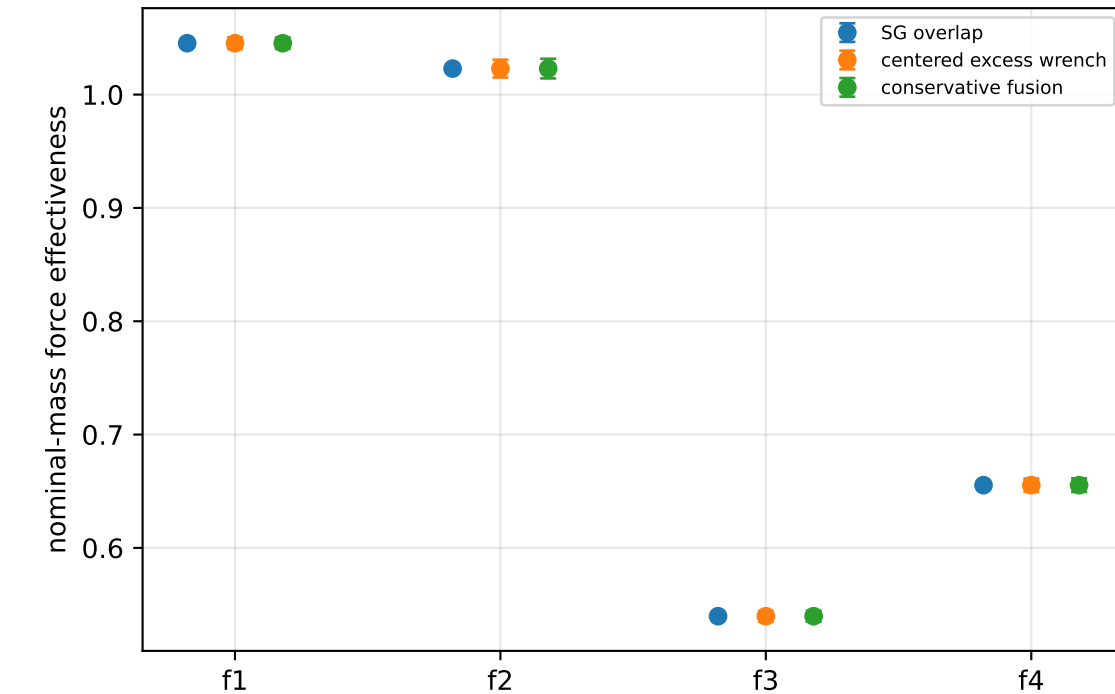
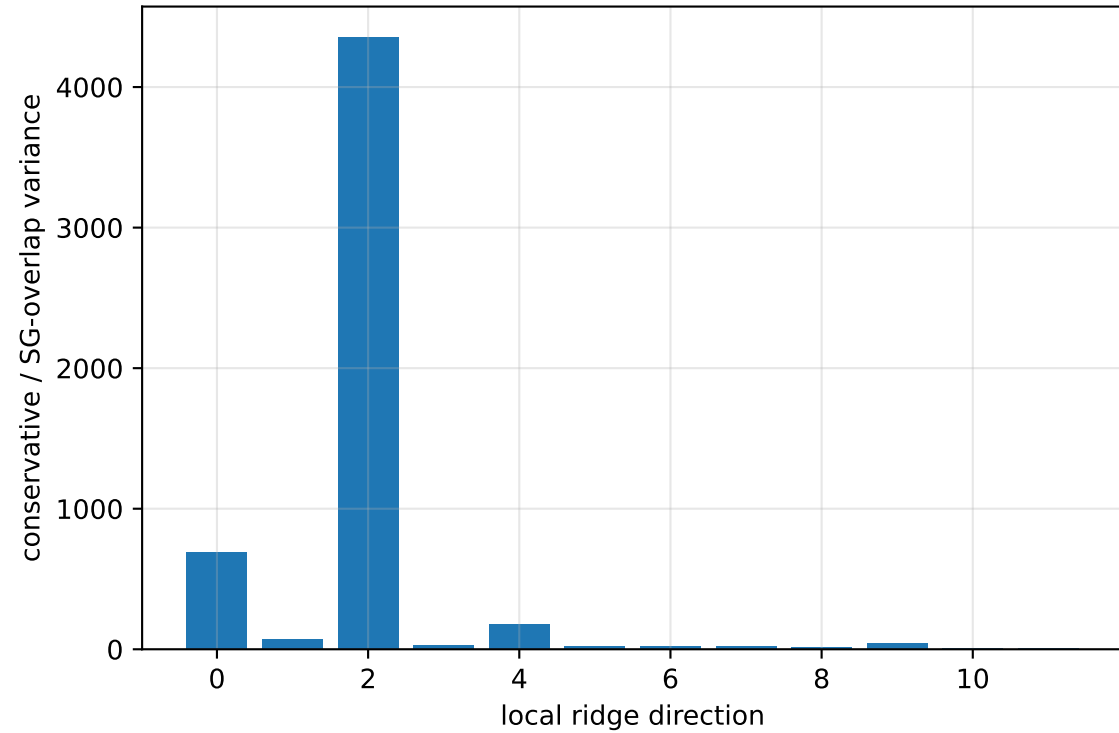
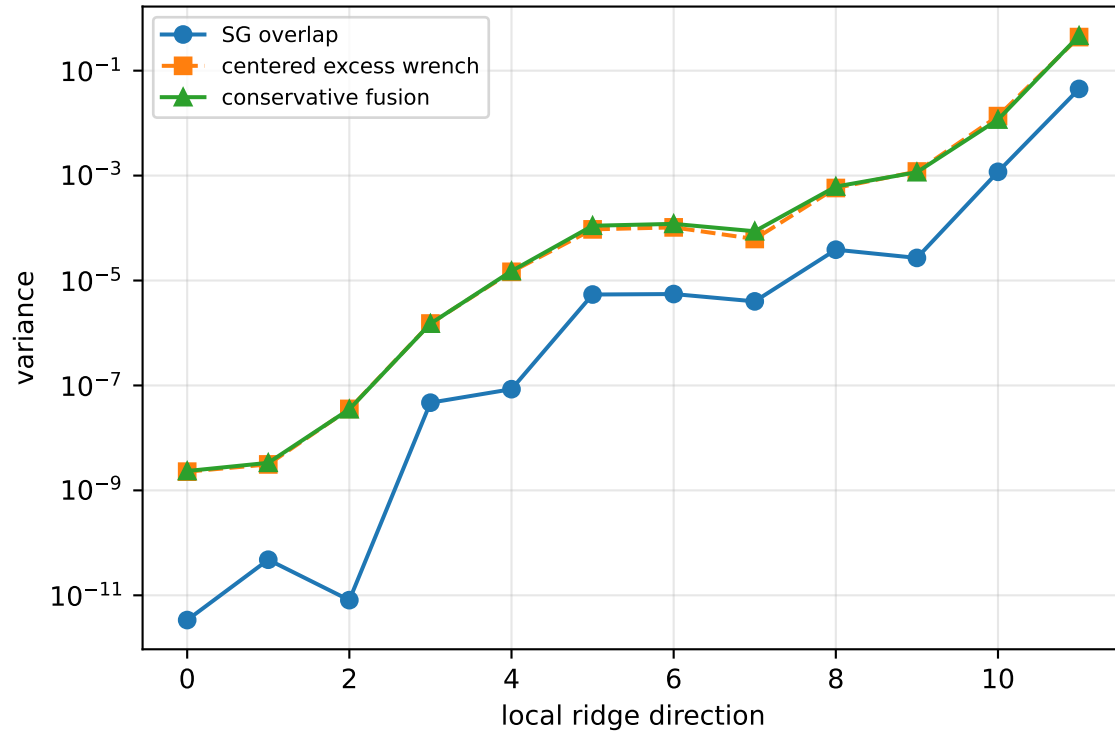
rotor lag cell: (0.175582647, 0.175611973] s

representative: 0.17559731 s

default: ridge spectrum (rank 13, nullity 1, $||jv||=7.94e-14$)



default: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 401.4$
wrench-to-acceleration recovery max error = $2.66\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

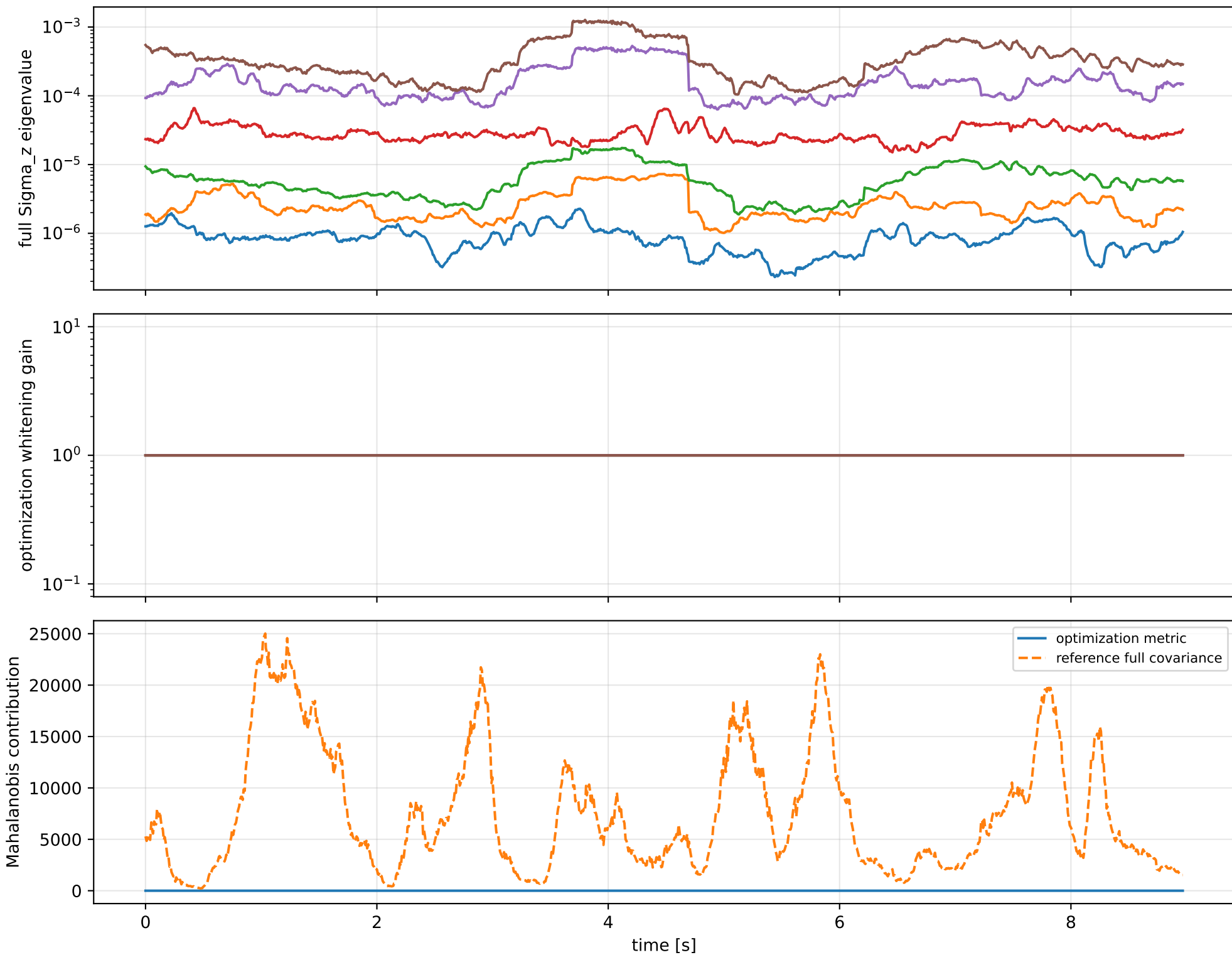
index 11: variance=0.4624, ratio=10.3
 $\log_{\text{mass}}=+0.00157$; second_moment_diag_1=+0.0653
second_moment_diag_2=-0.0608; second_moment_diag_3=-0.0128
second_moment_offdiag_12=+0.992; second_moment_offdiag_13=-0.0882
second_moment_offdiag_23=+0.0236; cog_x=-0.000556
cog_y=+0.000269; cog_z=-0.000373
log_force_eff_1=+0.00331; log_force_eff_2=-0.00243
log_force_eff_3=+0.0015; log_force_eff_4=+0.00438

index 10: variance=0.01168, ratio=9.901
 $\log_{\text{mass}}=-0.0185$; second_moment_diag_1=-0.0144
second_moment_diag_2=-0.0444; second_moment_diag_3=+0.142
second_moment_offdiag_12=+0.0887; second_moment_offdiag_13=+0.981
second_moment_offdiag_23=-0.0427; cog_x=+0.00582
cog_y=-0.000797; cog_z=-0.00063
log_force_eff_1=-0.0349; log_force_eff_2=+0.0056
log_force_eff_3=+0.0195; log_force_eff_4=-0.0546

index 9: variance=0.00116, ratio=43.03
 $\log_{\text{mass}}=-0.169$; second_moment_diag_1=+0.00103
second_moment_diag_2=+0.893; second_moment_diag_3=-0.0375
second_moment_offdiag_12=+0.0539; second_moment_offdiag_13=+0.0387
second_moment_offdiag_23=+0.216; cog_x=-0.00428
cog_y=-0.000447; cog_z=-0.0111
log_force_eff_1=-0.164; log_force_eff_2=-0.172
log_force_eff_3=-0.212; log_force_eff_4=-0.141

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

default: optimization vs reference covariance



default: wrench / closure (force max $1.58\text{e-}14$, torque max $1.11\text{e-}16$)

